

LIU HENGZHI

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EDUCATION

- National University of Singapore**2026 – June 2029
Bachelor of Computing in Computer Science (Tuition Grant)
- Temasek Polytechnic, Singapore**2023 – 2026
Diploma in Biomedical Engineering with Merit
 - Awarded Polytechnic Scholarship, Course Bronze Medalist, and Director's List honors.

SKILLS

Programming Languages: Python, C++, C, SQL
Systems & AI Infra: LLM Agents, System Design Basics, Embedded Systems, Real-time Systems (FreeRTOS), Firmware Development
Tools & Frameworks: Git/GitHub, Linux, FastAPI, MySQL, AWS (Aurora), CAD, Adobe Suite

TECHNICAL PROJECTS

- DTx Cognitive SaaS (ADHD Therapeutics)**2026
*Full-Stack Developer**React, FastAPI, AWS Aurora, LLM Agents*
 - Built a clinical ADHD SaaS platform featuring adaptive test pacing calibrated to severity.
 - Reduced data latency by implementing local telemetry batching with AWS token authentication.
 - Built an autonomous AI Copilot via Gemini API with local report caching.
- Smart Mahjong Play-Board**2026
*Independent Systems Developer**C++, ESP32, PCB Design, Embedded Firmware*
 - Developed ESP32-based smart play-board hardware featuring custom PCB circuit integration.
 - Achieved lag-free, real-time physical-to-digital state updates by writing optimized ESP32 C++ firmware.

WORK EXPERIENCE

- National University of Singapore (iHealthtech)**Mar 2025 – Jul 2025
*Research Assistant Intern**Singapore*
 - Generated high-quality experimental datasets for biomedical research by setting up microscale testing workflows.
 - Ensured data integrity across all testing pipelines by performing comprehensive validation checks on hardware instruments.
 - Improved data traceability by establishing structured digital documentation standards for all experimental records.

CO-CURRICULAR ACTIVITIES

- WorldSkills Singapore (Additive Manufacturing)**2024
 - Won 1st Place (TP Selection Round), demonstrating advanced CAD modeling and precision design.
- Electronic Design & Robotics Competitions**2024
 - Developed embedded logic systems and real-time control algorithms for robotics challenges.